

Configure Failover Routing Policy in Amazon Route 53

Summary:-

In this lab, a Failover Routing Policy is configured in Amazon Route 53 to provide an Active–Passive architecture. Two EC2 web servers are configured as primary and secondary endpoints. A Route 53 Health Check monitors the primary server, allowing Route 53 to automatically redirect traffic to the secondary server when the primary becomes unavailable and restore traffic when it recovers.

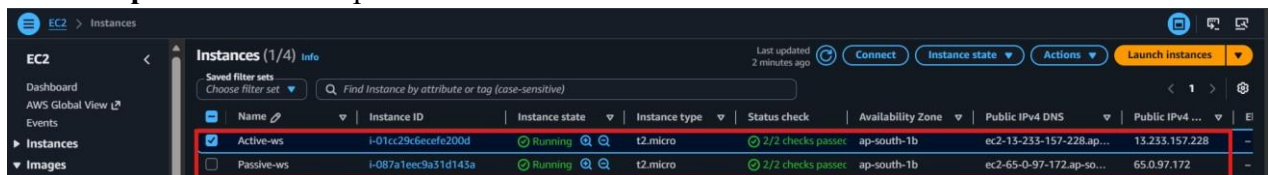
Let's Start:-

- Create a Route 53 Health Check for the primary web server.
- Configure Failover Routing Policy records.
- Associate the primary record with the Health Check.
- Simulate a primary server failure.
- Verify automatic failover and recovery.

Lab Steps:-

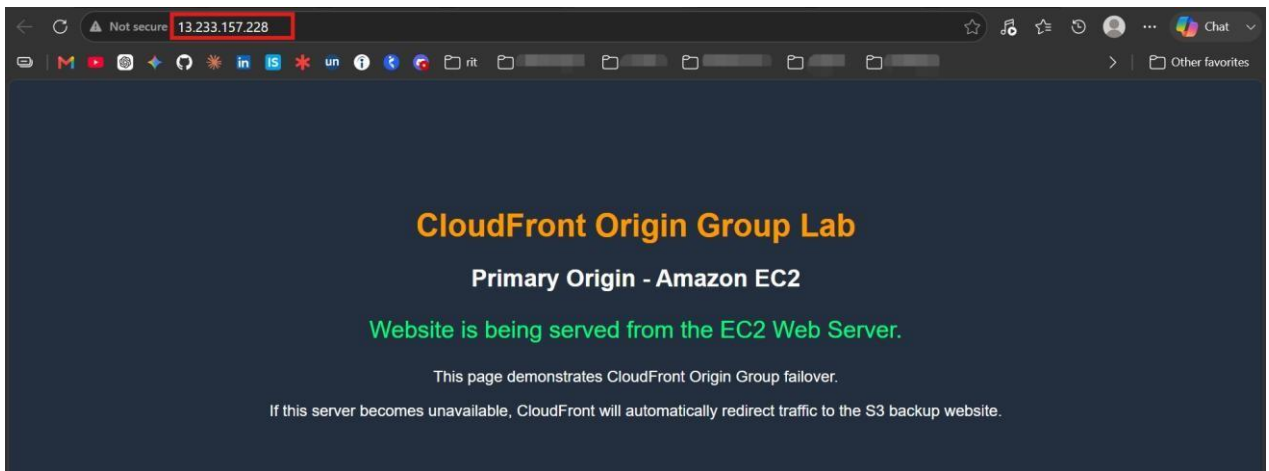
Step 1 – Create a Health Check

- **Note:** Before attempting this lab, complete Steps 1–2 of the Route53_DNS_Record_Types_AWS lab to configure Route 53 and host a website on an EC2 instance named **web-server-1**.
- **Repeat** the same steps to create another EC2 instance named **web-server-2**.

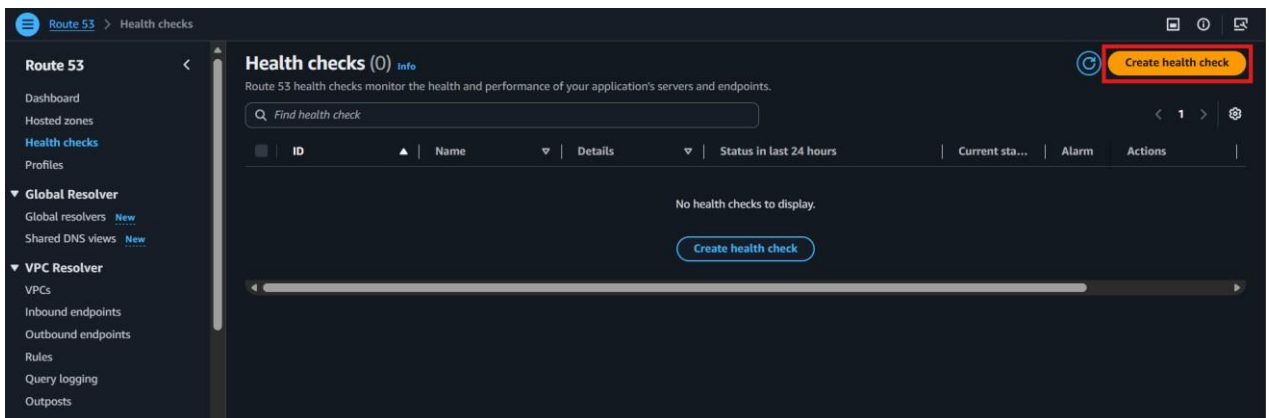


Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
Active-ws	i-01cc29c6eefe200d	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-13-233-157-228.ap...	13.233.157.228
Passive-ws	i-087a1eecd31d143a	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-65-0-97-172.ap-so...	65.0.97.172

- This step prepares the required Route 53 and EC2 resources for the Failover Routing Policy.



- Open the **Amazon Route 53** console.
- Select **Health Checks**.
- Click **Create Health Check**.



- Configure the following:
 - **Name:** Active_Web
 - **Protocol:** HTTP
 - **IP Address:** Public IPv4 address of the primary web server
 - **Port:** 80

Create health check Info

Create a new health check to monitor the health of a specified resource.

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#) ↗

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

Active-web

The name must have 1-256 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource

☒ **Endpoint**
Establish a connection with the resource to determine its health status.

☐ **Calculated health check**
The status of the health check is based on the status of the other health checks.

☐ **CloudWatch alarm**
The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ **IP address**

☐ **Domain name**

IP address
The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

HTTP 13.233.157.228:80

- Leave the remaining settings as default.
- Click **Create Health Check**.
- Wait until the Health Check status becomes **Healthy**.

Health checks (1) Info

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

Find health check

ID	Name	Details	Status in last 24 hours	Current s...	Alarm	Actions
dded12c2-0391-...	Active-web	http://13.233.1...		Healthy	None, ...	

- This step monitors the health of the primary web server.

Step 2 – Create the Primary Failover Record

- Open the **Hosted Zone**.
- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** Public IPv4 address of the primary web server
 - **TTL:** 60 Seconds
 - **Routing Policy:** Failover
 - **Failover Record Type:** Primary
 - **Health Check:** Active_Web
 - **Record ID:** Primary

Route 53 > Hosted zones > cloudservicesdemo.in > Create record

Record 1

Record name: **learn** .cloudservicesdemo.in

Record type: **A – Routes traffic to an IPv4 address and some AWS resources**

Value: **13.233.157.228**

TTL (seconds): **60** (1m, 1h, 1d)

Routing policy: **Failover**

Failover record type: **Primary**

Health check ID: **dded12c2-0391-467f-9b8a-9011a33479b7**

Record ID: **Primary**

- Click **Create Records**.
- This step creates the active DNS record monitored by the Health Check.

Step 3 – Create the Secondary Failover Record

- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** Public IPv4 address of the secondary web server
 - **TTL:** 60 Seconds
 - **Routing Policy:** Failover
 - **Failover Record Type:** Secondary
 - **Record ID:** Secondary

Route 53 > Hosted zones > cloudservicesdemo.in > Create record

Record 1

Record name: **learn** .cloudservicesdemo.in

Record type: **A – Routes traffic to an IPv4 address and some AWS resources**

Value: **65.0.97.172**

TTL (seconds): **60** (1m, 1h, 1d)

Routing policy: **Failover**

Failover record type: **Secondary**

Health check ID - optional: **Choose health check**

Record ID: **Secondary**

- Click **Create Records**.
- This step creates the standby DNS record used during failover.

Step 4 – Verify the Primary Server

- Open **Command Prompt**.
- **Run:**
 - `nslookup learn.<your-domain-name>`
- Verify that the returned IP address belongs to the primary web server.

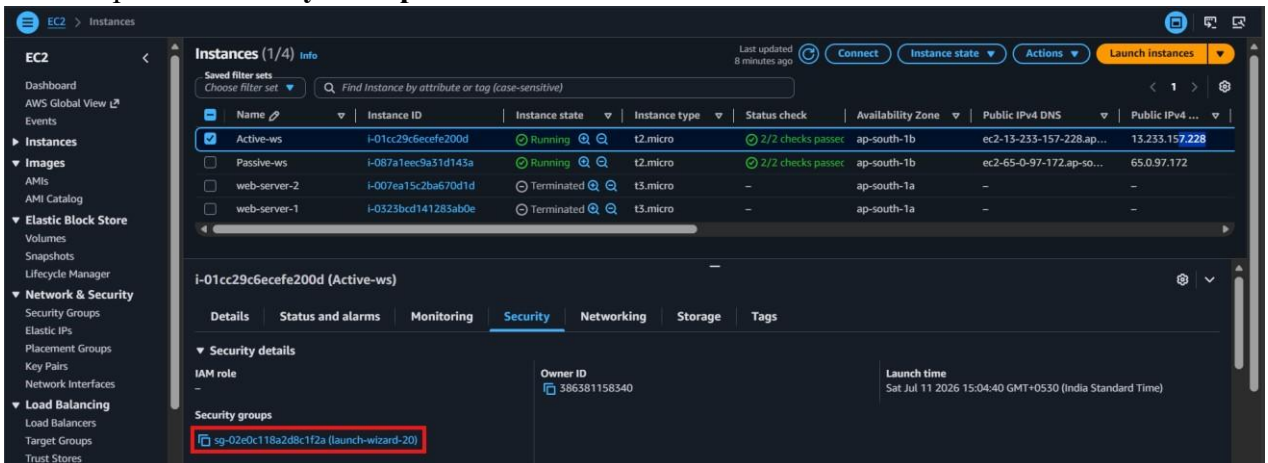
```
PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server:      reliance.reliance
Address:     2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:   learn.cloudservicesdemo.in
Address: 13.233.157.228
PS C:\Users\HP\downloads> |
```

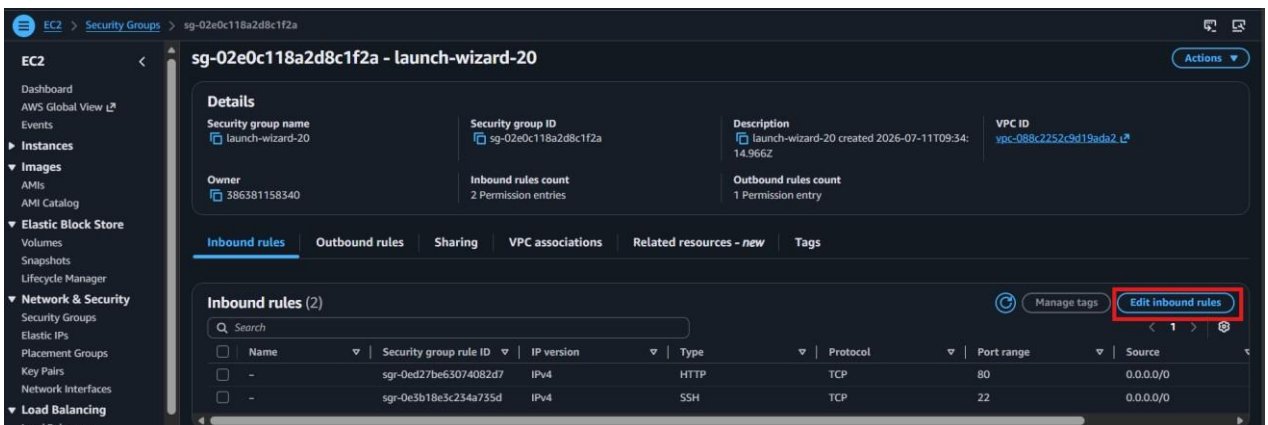
- This step confirms that Route 53 resolves the primary endpoint while it is healthy.

Step 5 – Simulate a Primary Server Failure

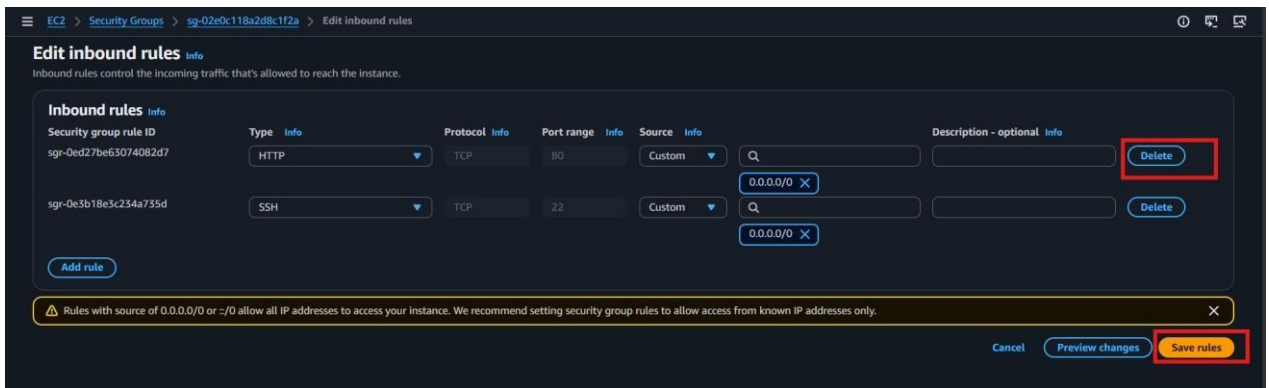
- Open the **EC2** console.
- Select the primary web server.
- Open its **Security Group**.



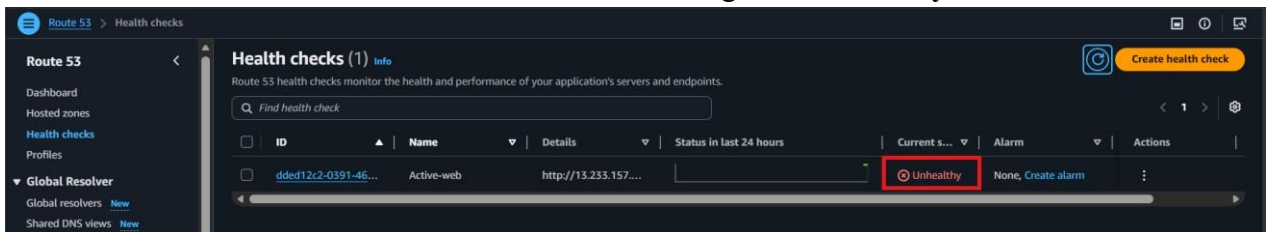
- Edit the inbound rules.



- Remove the **HTTP (Port 80)** rule.



- Save the changes.
- Wait until the Route 53 Health Check status changes to **Unhealthy**.



- This step simulates a failure of the primary web server.

Step 6 – Verify Automatic Failover

- Open **Command Prompt**.
- Run:
 - `nslookup learn.<your-domain-name>`
- Wait for the Health Check and DNS TTL to expire.

```
PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server: reliance.reliance
Address: 2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:   learn.cloudservicesdemo.in
Address: 13.233.157.228

PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server: reliance.reliance
Address: 2405:201:681d:ea79::c0a8:1d01

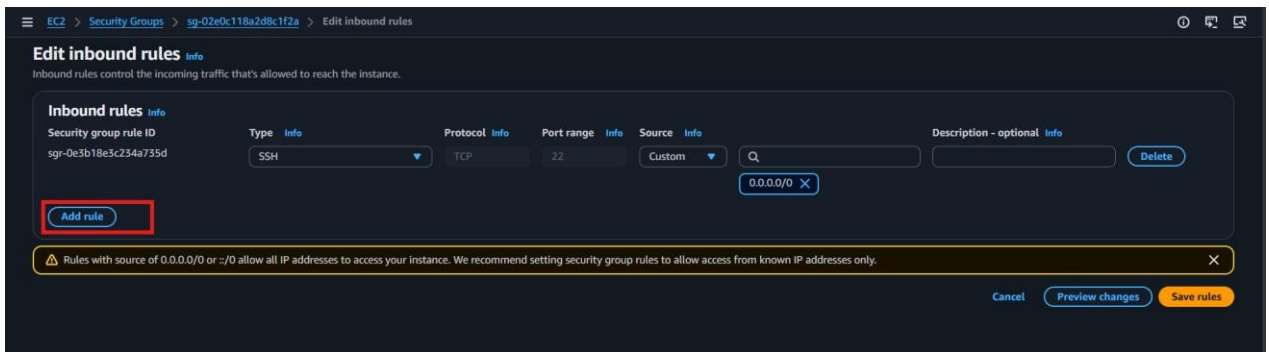
Non-authoritative answer:
Name:   learn.cloudservicesdemo.in
Address: 65.0.97.172

PS C:\Users\HP\downloads> |
```

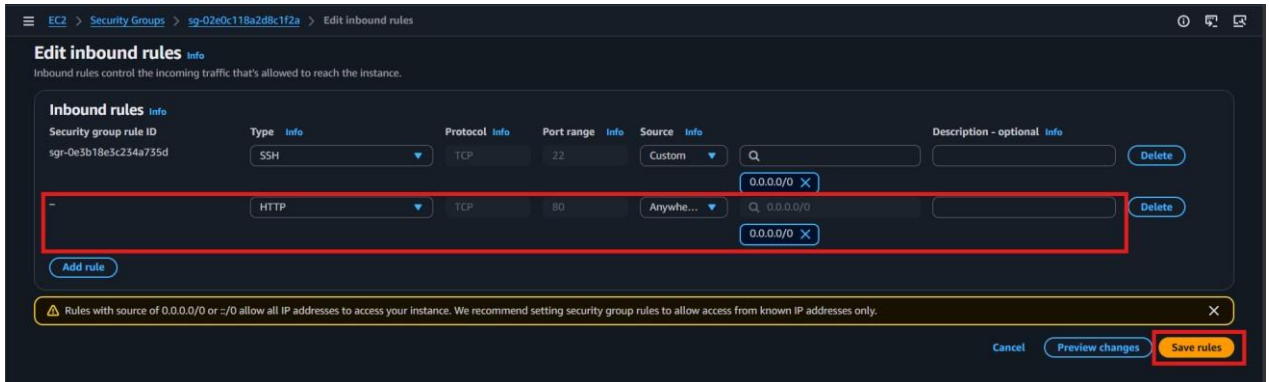
- Verify that the returned IP address belongs to the secondary web server.
- This step verifies that Route 53 automatically redirects traffic to the standby server.

Step 7 – Restore the Primary Server

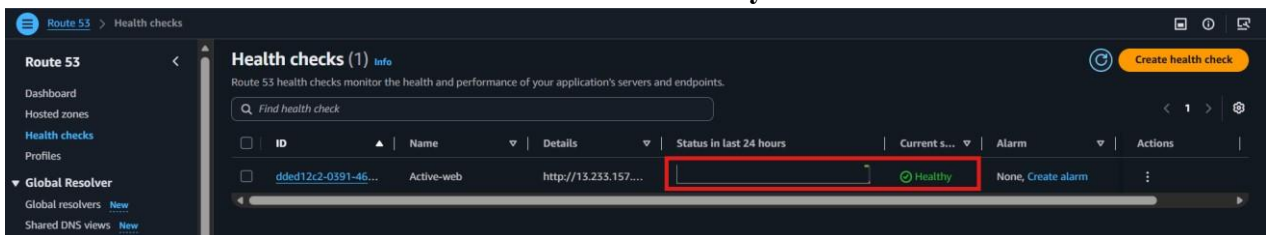
- Edit the Security Group of the primary web server.
- Add the **HTTP (Port 80)** inbound rule.



- Save the changes.



- Wait until the Health Check status becomes **Healthy**.



- This step restores the primary web server.

Step 8 – Verify Automatic Recovery

- Open Command Prompt.
- Run:
 - nslookup learn.<your-domain-name>
- Wait for the DNS TTL to expire.


```
PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address: 13.233.157.228

PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
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Address: 65.0.97.172

PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address: 13.233.157.228

PS C:\Users\HP\downloads>
```

- Verify that the returned IP address belongs to the primary web server.
- This step confirms that Route 53 automatically switches traffic back to the primary server.

Verification

- Successfully created a Route 53 Health Check.
- Configured Primary and Secondary Failover Routing records.
- Verified DNS resolution to the primary web server.
- Simulated a primary server failure and confirmed automatic failover.
- Restored the primary server and verified automatic recovery.